Kun-Ying Lee

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Robot Learning | Humanoid Locomotion | Motion Generation

Education

Tsinghua University 2025–2029 (expected)
Undergraduate, Yao Class GPA: 3.8/4.0
Computer Science (Artificial Intelligence Track)
Institute for Interdisciplinary Information Sciences (IIIS)

Selected Publications

OMG: Omni-Modal Motion Generation for Generalist Humanoid Control
Siqiao Huang*, **Kun-Ying Lee***, Dongming Qiao*, Guanqi He*, Zhenyu Wang, Yitang Li, Shaoting Zhu, and Hang Zhao.
arXiv:2606.10340, 2026. Paper | Project | Code
RSS 2026 workshops: ExWBC (**Oral**); RoboData (**Spotlight**).

Hiking in the Wild: A Scalable Perceptive Parkour Framework for Humanoids
Shaoting Zhu*, Ziwen Zhuang*, Mengjie Zhao, **Kun-Ying Lee**, and Hang Zhao.
Conference on Robot Learning (CoRL), 2026 (accepted). Paper | Project | Code

The Indefinite Integral of the *n*th Power of Inverse Trigonometric Functions
Kun-Ying Lee.
Mathmedia, Academia Sinica, 49(2):91–101, 2025. Article

* Equal contribution.

Research Experience

MARS Lab, Tsinghua University 2025–present
Research Assistant | Advisor: Prof. Hang Zhao

- **OMG-Posttrain.** Developing reinforcement learning methods for post-training OMG, focusing on closed-loop execution of highly dynamic motions with asynchronous motion generation and control.
- **OMG.** Developed a scalable, omni-modal humanoid foundation model that maps language, audio, and human reference motions to executable whole-body motions on Unitree G1.
- **Hiking in the Wild.** Validated sim-to-sim policy transfer and assisted with real-robot deployment of a depth-conditioned humanoid parkour policy.

Leadership & Robotics

TH-MOS Robot Soccer Team, Tsinghua University Sept. 2025–present
Role: Gait Control Group Lead

- Lead soccer-skill development and train policies for omnidirectional velocity tracking, passing, shooting, goalkeeping, and fall recovery.
- Built a **MuJoCo**-based RoboCup simulation environment integrating decision-making, locomotion control, and perception for rapid policy evaluation and iteration.

Selected Honors & Awards

5th Place, RoboCup 2026 Incheon (TH-MOS team) 2026
2nd Place, RoboCup Asia-Pacific Tianjin Invitational Tournament (550W team) 2026
3rd Place, RoboCup China Open (TH-MOS team) 2026
Leading Talent (Lingjun) Award, Tsinghua University 2025